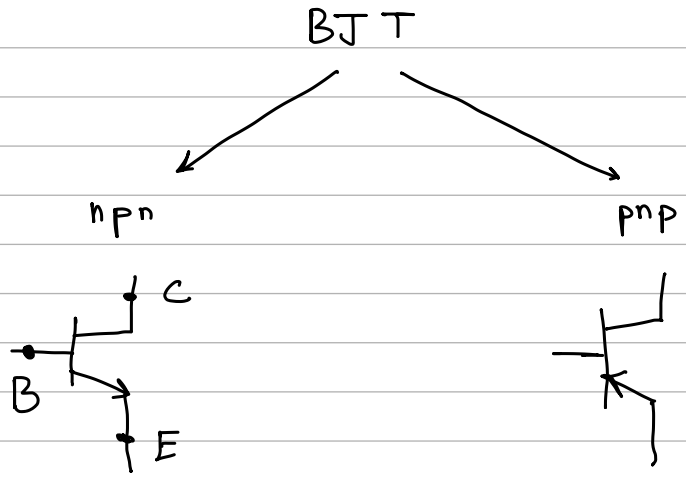
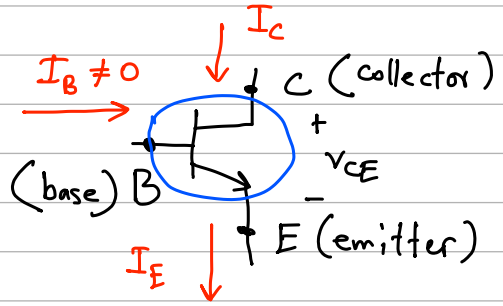
# BJT (Bipolar Junction transistor):




→ mainly will deal with npn.

## BJT




$$I_B + I_C = I_E$$ ✓

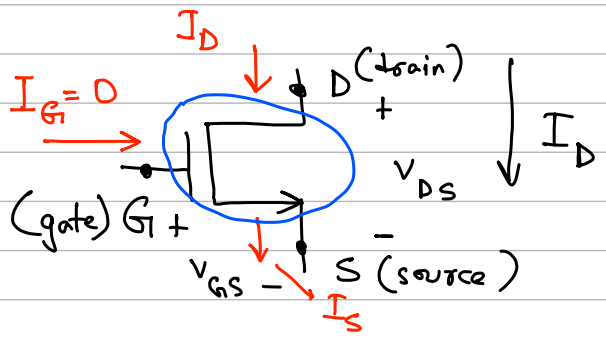
$$I_C \neq I_E$$

IV characteristics ($I_C$ vs. $V_{CE}$):

→ $I_B$ controls this graph

→ current controlled

## MOSFET




$V_{GS} < V_T$ : cutoff

$V_{GS} > V_T$ : triode or saturation

↓ ↓

$V_{DS} \leq V_{ov}$ $V_{DS} > V_{ov}$

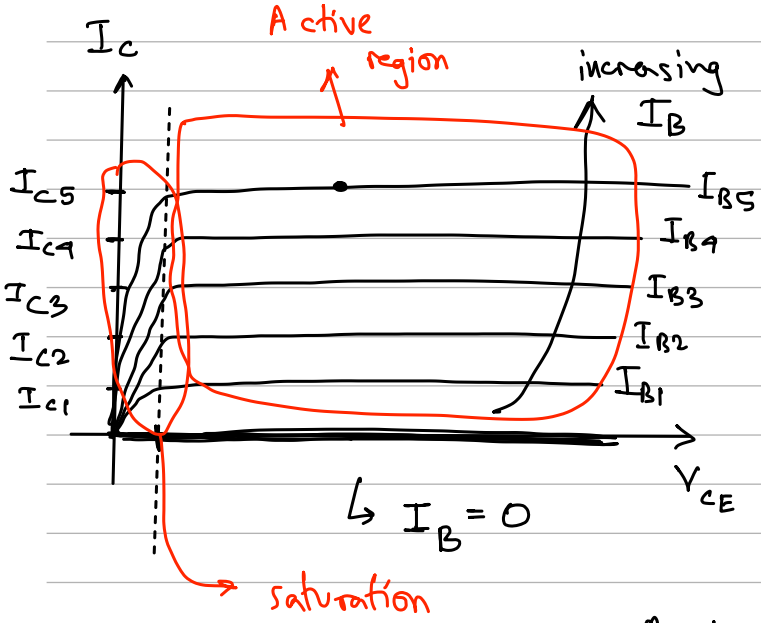
$$V_{ov} = V_{GS} - V_T$$

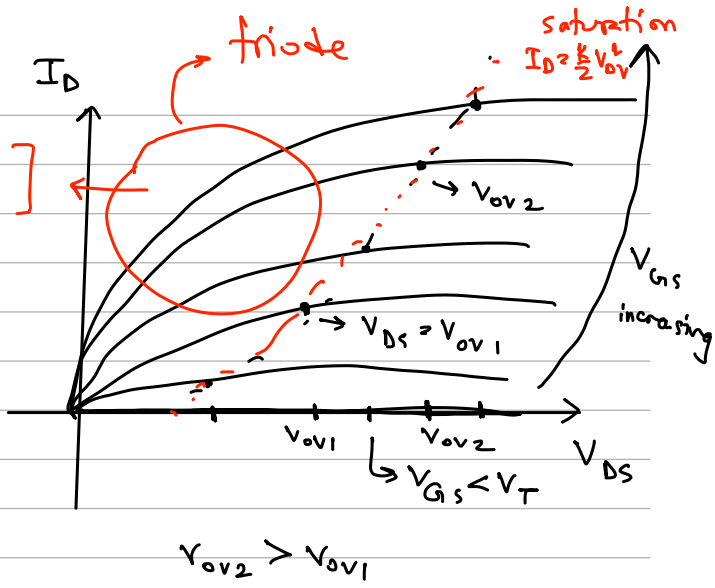
$I_G = 0$, $I_D = I_S$

IV characteristics ($I_D$ vs $V_{DS}$):

→ $V_{GS}$ controls this graph

→ voltage controlled

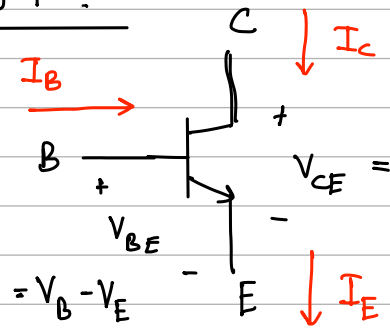



## Analogue:

| BJT | MOSFET |
|---|---|
| cutoff | cutoff |
| saturation (much easier than MOSFET) | triode |
| active | saturation |

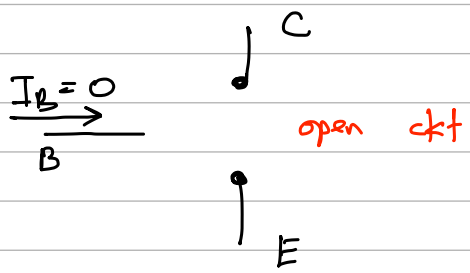
## S-model of BJT:




$$I_E = I_B + I_C$$

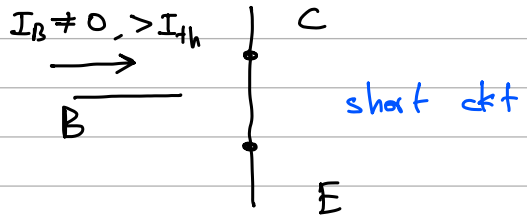
* Current controlled ($I_B$) switch.
* Interested in $I_C$, $V_{CE}$

$I_B = 0$

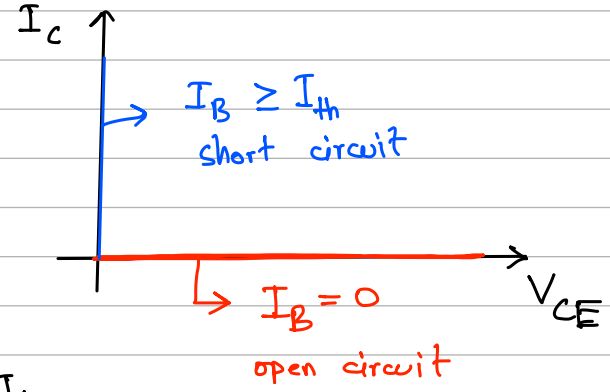


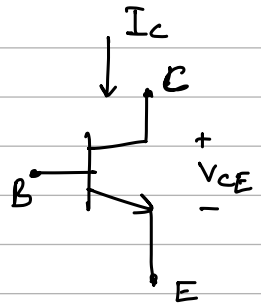

W OFF ( logicstate 0)

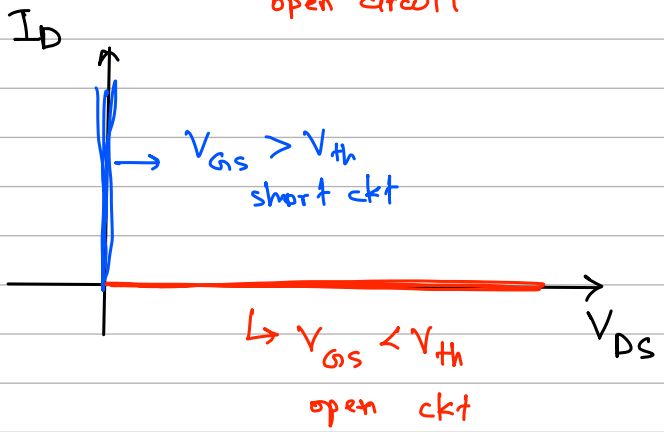
$I_B \geq I_{th}$

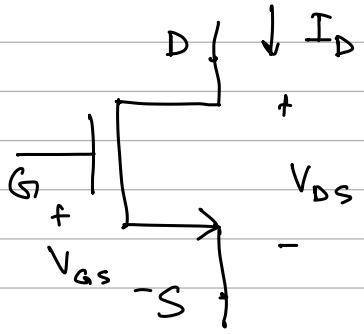



W ON ( logicstate 1)



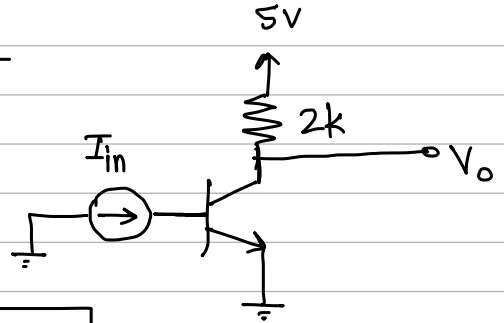




* What if $0 < I_B$ and $I_B < I_{th}$ ?

→ Invalid in S-model

→ $0 < I_B < I_{th}$ : actually 'active mode' in BJT

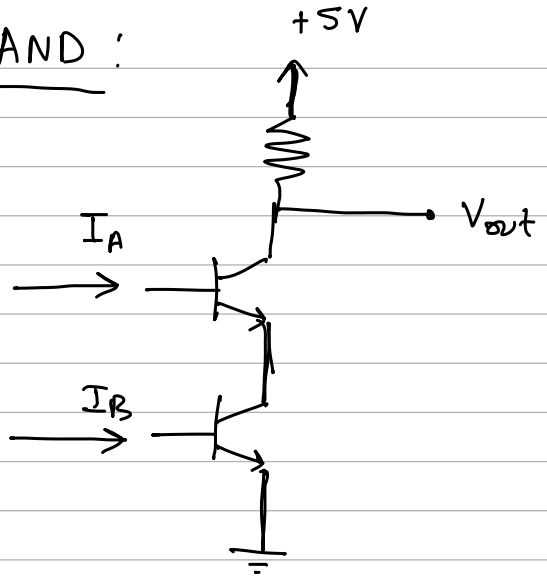
NOT inverter !




| $I_{in}$ | $V_o$ |
|---|---|
| 0mA (0) | 5V (1) |
| $I_{th}$ (1) | 0V (0) |

Output = $\overline{\text{Input}}$

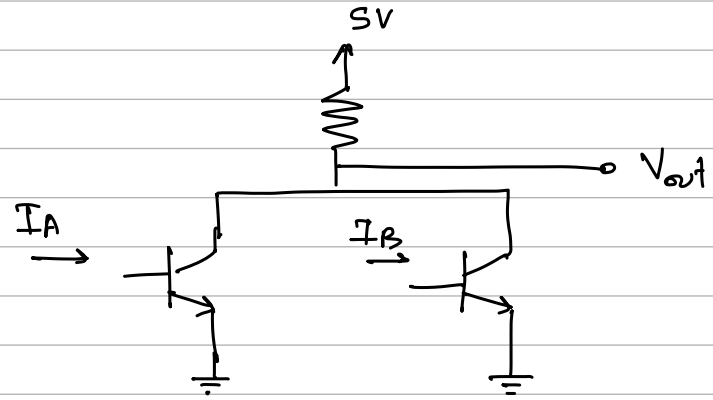
## NAND :




$out = \overline{AB}$

| $I_A$ | $I_B$ | $V_{out}$ |
|---|---|---|
| 0mA (0) | 0mA (0) | 5V (1) |
| 0mA (0) | $I_{th}$ (1) | 5V (1) |
| $I_{th}$ (1) | 0mA (0) | 5V (1) |
| $I_{th}$ (1) | $I_{th}$ (1) | 0V (0) |

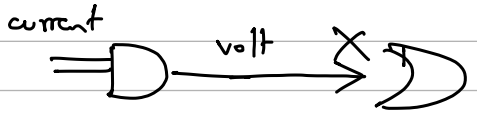
## NOR :



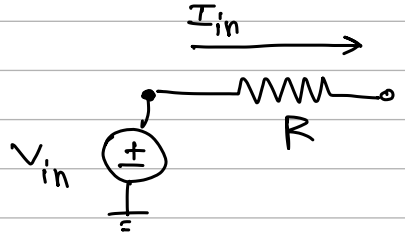

$out = \overline{A+B}$

## Problem :

① Cascading issue [ gate output voltage, but input current ]




* Can be addressed by using voltage source to create base current.

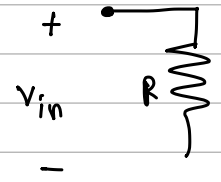



② Current source ?

③ current setting to zero is tough

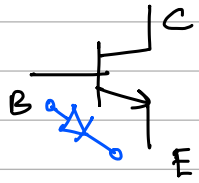
[ for mosfet :
$V_{GS} < V_{th}$ → logic 0
for BJT :
$I_B = 0$ → logic 0 ]

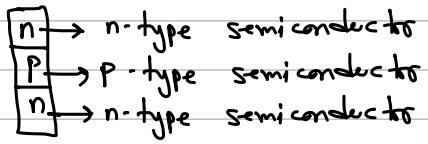
# Voltage control!



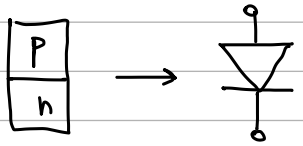

$i = \frac{V_{in}}{R} \Rightarrow V = Ri$

→ we use resistors to convert from voltage to current



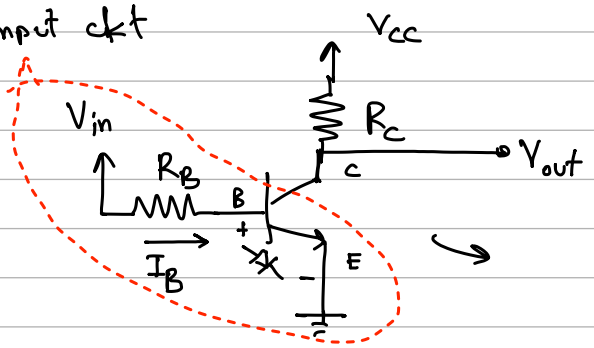




bet$^n$ Base and Emitter!



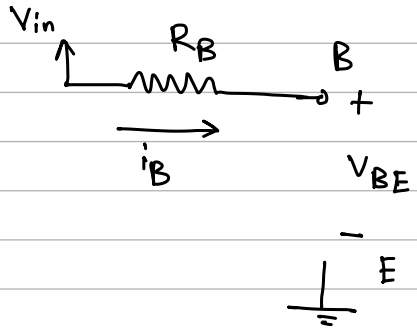

## E.g: Inverter:

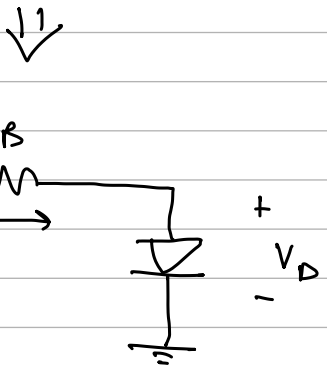
input ckt




| $I_B$ | $V_{in}$ | $V_{out}$ |
|---|---|---|
| $I_B = 0mA$ (0) | $V_{in} < 0.7V$ | 5V (1) |
| $I_B \geq I_{th}$ (1) | $V_{in} > V_{th}$ | 0V (0) |

input (control) circuit:




## $V_{in} < 0.7$:

→ BE is not forward biased

[B ⊘ E is 'OFF']

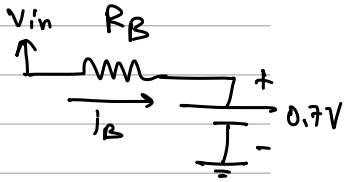
→ $I_B = 0$.

## $V_{in} \geq 0.7$:

BE forward biased

$I_B > 0$.

when diode ON:




KVL:

$i_B R_B + 0.7 = V_{in} - 0$

or, $\boxed{i_B = \frac{V_{in} - 0.7}{R_B}}$

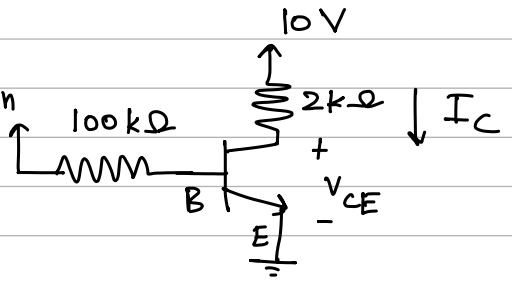
When $i_B = I_{th}$ :

$$I_{th} = \frac{V_{th} - 0.7}{R_B} \Rightarrow V_{th} = (I_{th} R_B + 0.7)\ V$$

When, $I_B = I_{th}$ , $V_{in} = V_{th}$ .

$\therefore V_{in} \geq V_{th}$ , $I_B \geq I_{th}$ .

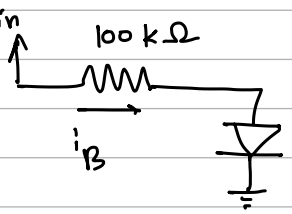
Eg :




$I_{th} = 0.2$ mA

find threshold of $V_{in}$ for which $I_B \geq I_{th}$ .

input ckt'.

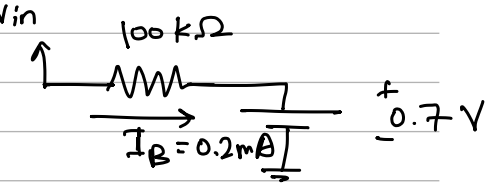



$i_B = \frac{V_{in} - V_D}{100\,k\Omega}$ : when diode ON $\rightarrow$ $i_B = \frac{V_{in} - 0.7}{100\,k\Omega}$

$\therefore V_{in} < 0.7V$ : Diode OFF, BE junction reverse biased.

$\therefore I_B = 0$ when $V_{in} < 0.7V$

$I_B = I_{th} = 0.2$ mA :



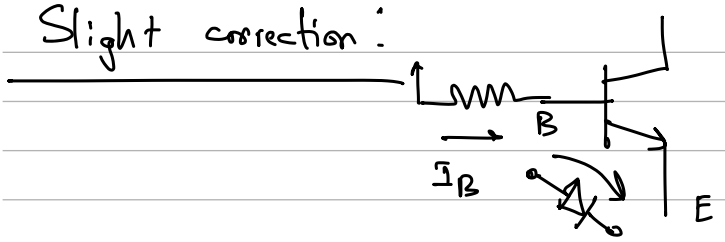

$0.2\ mA \times 100\ k\Omega = V_{in} - 0.8V$

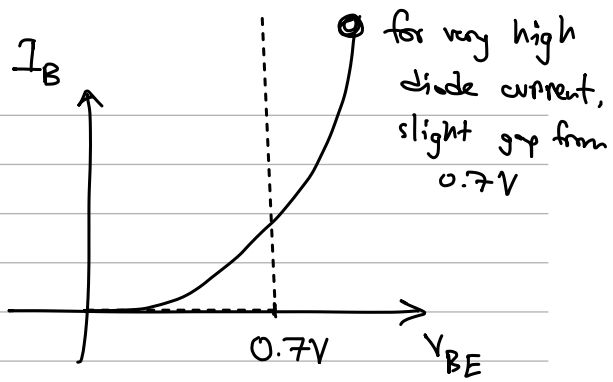
$V_{th} \leftarrow (V_{in}) = 20 + 0.8 = 20.8\ V$

$V_{in} = 20.8V$ , $I_B = 0.2$ mA $\Rightarrow V_{in} \geq \underline{20.8V}$ , $I_B \geq I_{th}$ .

$V_{th}$

Slight correction:

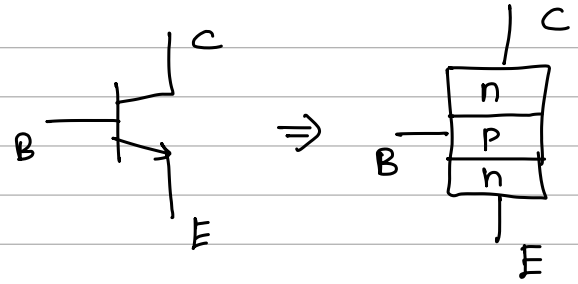


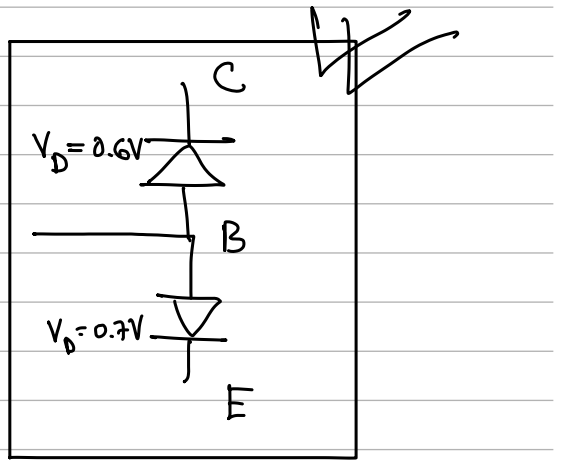




then for $I_B \geq I_{th}$ :

→ very high current, so diode drop assumed slightly more than 0.7V. ⟹ 0.8V

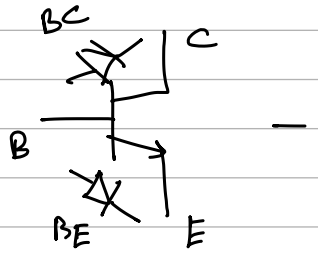
Physical structure:




→ not actual BJT structure

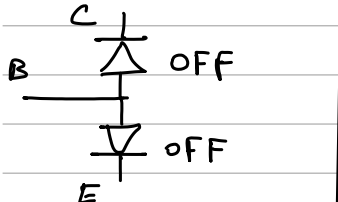
→ only used for analysis simplification




BE → ON, OFF
BC → ON, OFF
} 4 combinations, 4 modes of operation

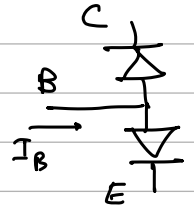
[cutoff] | [saturation]

BE 'OFF', BC 'OFF'



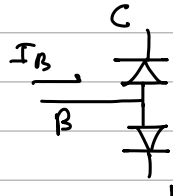

→ $I_B = 0$.
→ $I_C = 0$
$V_{BC} < 0.6$
$V_{BE} < 0.7$

BE 'OFF', BC 'ON' (reverse active)

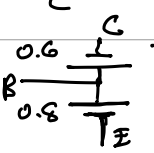



→ similar to 'active'
→ will not deal with this much

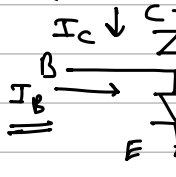
BE 'ON', BC 'ON'



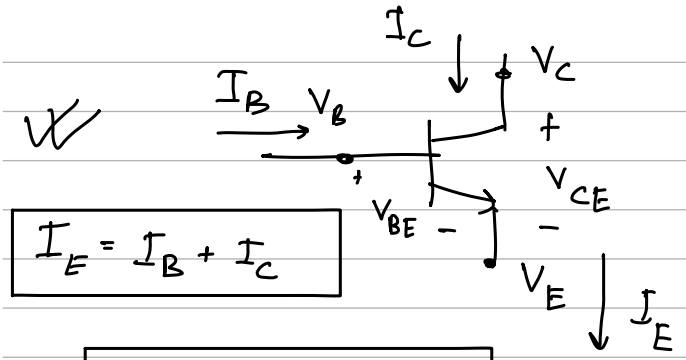

→ $I_B$ ↗
→ $I_C$ ↗




BE 'ON' BC 'OFF' (active)




→ $I_B$ ↗ $V_{BE} = 0.7V$
→ $I_C$ const, depends on $I_B$
$I_C \propto I_B \Rightarrow I_C = \beta I_B$




$I_E = I_B + I_C$

$I_B$ input

$I_C$ output

$V_{CE}$ used in x-axis of $I_C$ vs $V_{CE}$ (output characteristic graph)

$V_{BE} = V_B - V_E$

$V_{BC} = V_B - V_C$

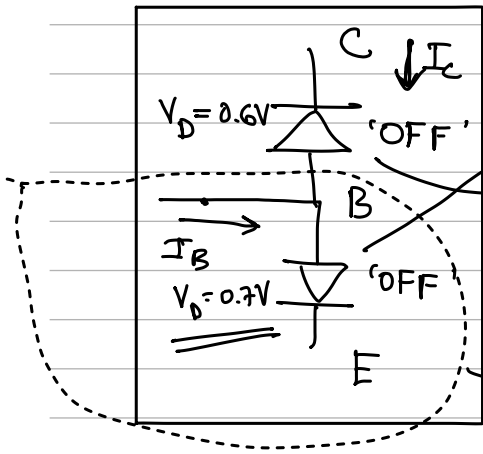
$V_{CE} = V_C - V_E$

$V_{BC} = V_B - V_E + V_E - V_C$

$= (V_B - V_E) - (V_C - V_E)$

$V_{BC} = V_{BE} - V_{CE}$

## ① Cutoff :



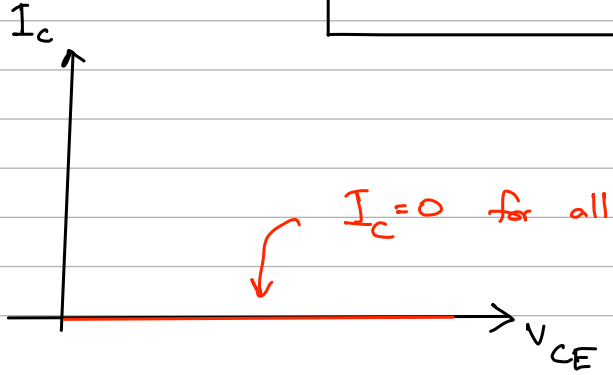

$V_{BE} < 0.7\,V$ → BE diode 'OFF'

$V_{BC} < 0.6\,V$ → BC diode 'OFF'

Consequence :

→ Input circuit current cannot be nonzero if BE OFF.

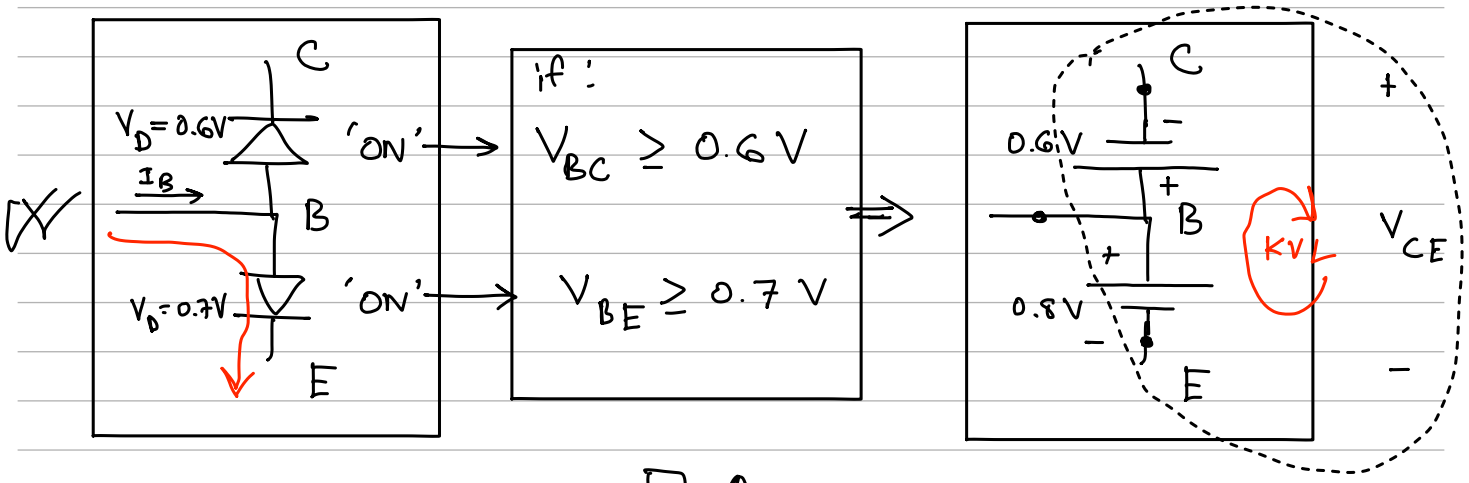
$\therefore I_B = 0,\ I_C = 0,\ I_E = 0$




$I_C = 0$ for all $V_{CE}$ in this mode

'open ckt'

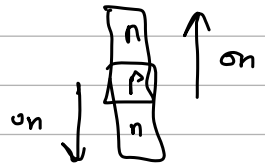
## ② Saturation : BE and BC junction forward biased and $I_B$ very high, so consider $V_{BE} \approx 0.8\,V$




If in saturation, then :

→ $V_{CE} = 0.2V$

→ Both junction forward biased

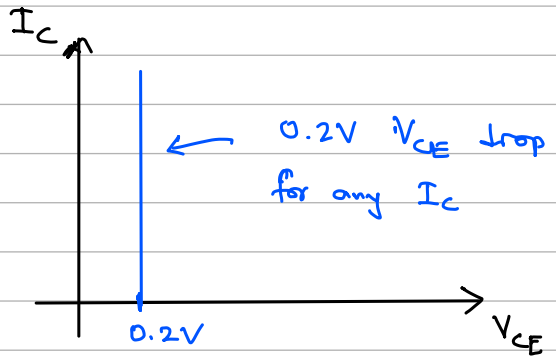



$$-V_{BE} + V_{BC} + V_{CE} = 0$$

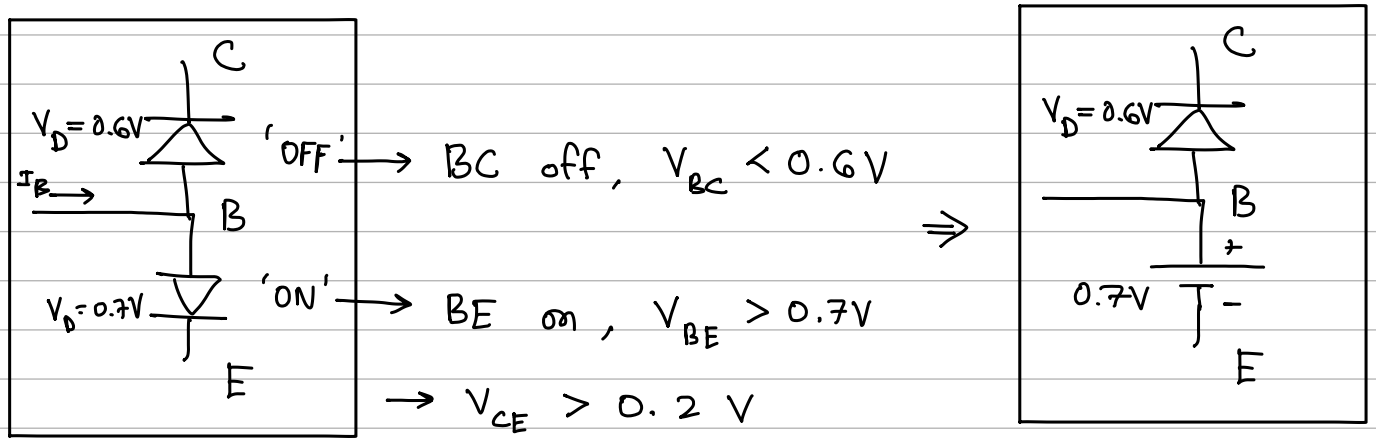
or, $V_{CE} = V_{BE} - V_{BC}$

$= 0.8 - 0.6$

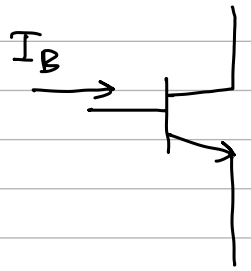
$= 0.2\,V$




## ③ Active : $I_C$ current saturates (and only depends on $I_B$).




BC is not 'open ckt'-ed. In this config., huge $I_C$ flows from C to B that depends only on $I_B$.



$I_C = \beta I_B$ — ① where $\beta$ = constant that is commonly called current gain

$$I_E = I_B + I_C \quad — ②$$
$$= I_B + \beta I_B$$
$$= (\beta+1) I_B$$

$$\beta = \frac{I_C}{I_B}$$

common values: (100), 150

$$I_B = \frac{1}{\beta} I_C \xrightarrow{②} I_E = \frac{1}{\beta} I_C + I_C = I_C \left(\frac{1}{\beta} + 1\right) = I_C \left(\frac{\beta+1}{\beta}\right)$$

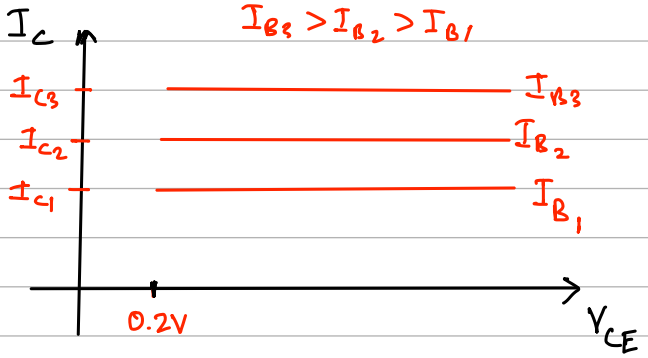
$$\therefore I_C = \left(\frac{\beta}{\beta+1}\right) I_E$$

$I_C = \beta I_B$ → current gain [collector - base]

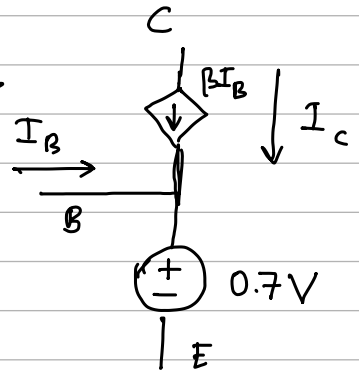
$$\Rightarrow I_C = \alpha I_E$$

→ current gain [collector - emitter]

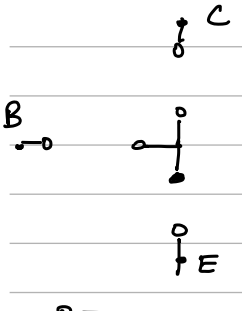
→ Behaves like a current source depending on input current $I_B$.




Active :

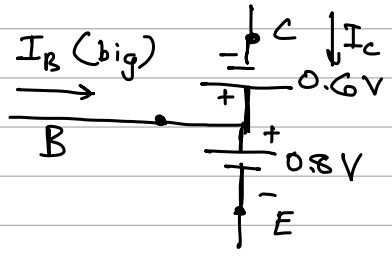



Cutoff :

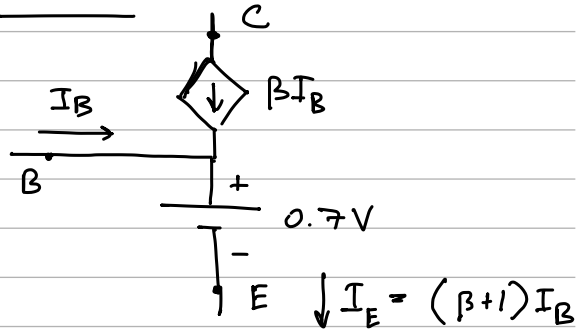



BJT OFF

Saturation :




Active :




$I_E = (\beta+1) I_B$

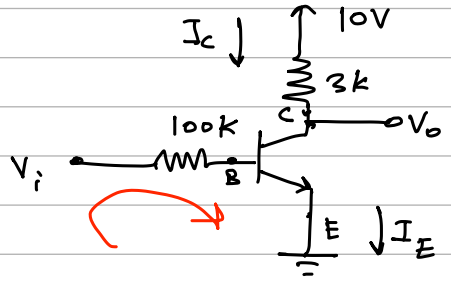
# Method of Assumed State :

① Assume
- Active [ $V_{BE} = 0.7V$, $I_C = \beta I_B$ ]
- Saturation [ $V_{BE} = 0.8V$, $V_{BC} = 0.6V$, $V_{CE} = 0.2V$ ]
- Cutoff [ $I_B = I_C = I_E = 0$ ]

② Solve [ KCL, KVL, node ]

③ Verify
- Active [ $V_{CE} > 0.2V$ ]
- Saturation [ $\frac{I_C}{I_B} < \beta$ ]
- Cutoff [ $V_{BE} < 0.7V$, and $V_{BC} < 0.6V$ ]

**Ex-1 :**



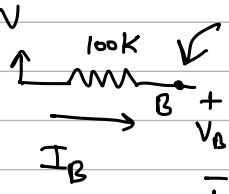

Find $v_o$ for $v_i = 1V$

$\beta = 100$

**Assume :** Active – ① $V_{BE} = 0.7V$

② $I_C = \beta I_B$

**Solve :**




$$\Rightarrow I_B = \frac{1 - 0.7}{100k} = 0.003\ mA$$

$I_C = \beta I_B$

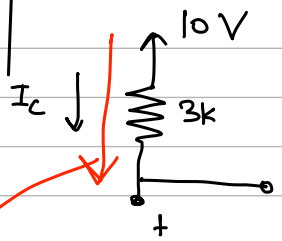
$= 100 \times 0.003\ mA$

$= 0.3\ mA$ ✓

$I_E = I_B + I_C = I_B + \beta I_B = (\beta+1) I_B$

$= 0.303\ mA$

$V_o = V_C$ here.




KLL

$I_C \times 3k = 10 - V_C$

or, $V_C = 10 - I_C \times 3k$

$V_o = 10 - I_C \times 3k$

$= 10 - .3 \times 3$

$= 10 - 0.9$

$= 9.1V$

Active : $V_{CE} > 0.2V$

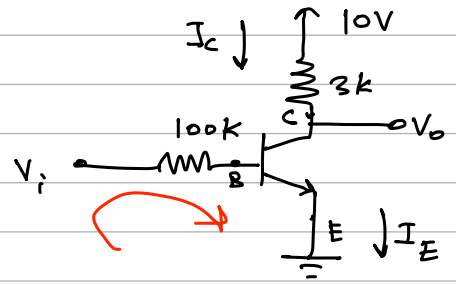
$V_C = 9.1V$, $V_E = 0$, $V_B = V_{BE} + V_E = 0.7V$

Verify :

$$V_{CE} = V_C - V_E = 9.1 - 0 = 9.1\,V > 0.2\,V.$$

∴ Correct assumption

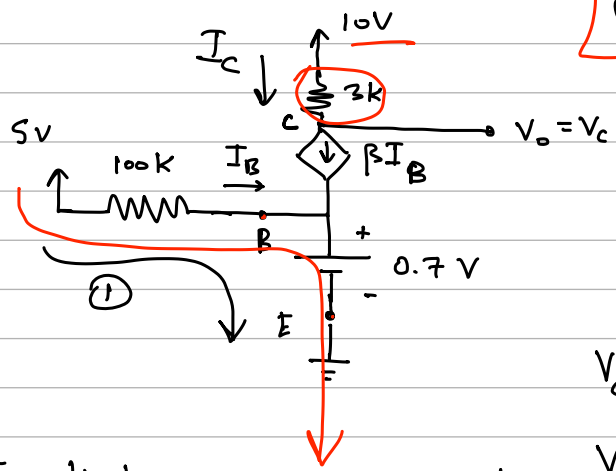
Ex-2 :




Find $v_o$ for $v_i = 5V$

$\beta = 100$

Assume : Active




① → $100 \times I_B + 0.7 = 5 - 0$

$$I_B = \frac{5 - 0.7}{100} = 0.043\ mA$$

$$I_C = \beta \times I_B = 100 \times 0.043 = 4.3\ mA$$

$$V_C = 10 - I_C \times 3k = -2.9\,V$$

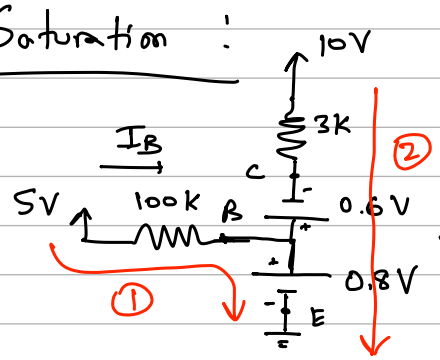
$V_E = 0\,V$

$$V_{CE} = V_C - V_E = -2.9 < 0.2\,V$$

BE diode ON. Then two states possible : ① Active ② Saturation

∴ Assumption wrong! Not in active state

Saturation :




① → $100\,I_B + 0.8 = 5 - 0$

$$I_B = \frac{5 - 0.8}{100} = 0.042\ mA$$

② → $3I_C - (0.6 + 0.8) = 10 - 0$ (+0.2V)

$$I_C = \frac{10 - 0.8 + 0.6}{3}$$

$3I_C + V_{CE} = 10 - 0$

$$= \frac{9.8}{3} = 3.29\ mA$$

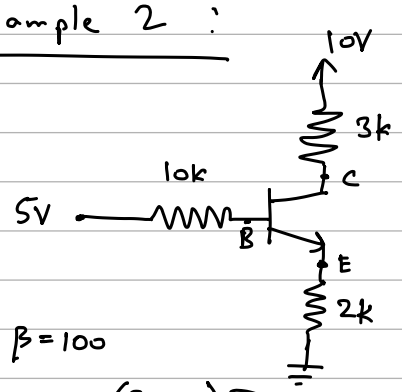
No formula for $I_C$ now.

Use output KVL line eqn

$$\frac{I_C}{I_B} = \frac{3.29}{0.042} = 78.33 < 100$$

[ Normally, $\beta_{saturation} < \beta_{active}$ ]

## Example 2 :




$\beta = 100$

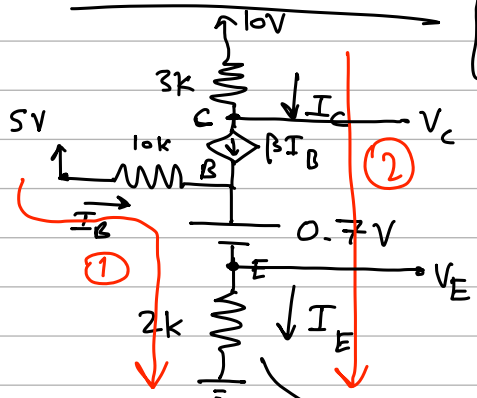
Find $I_B$, $I_C$, $I_E$, $V_{CE}$

(1) Assume : Active

$I_E = I_B + I_C = I_B + \beta I_B$

$I_E = (\beta+1) I_B$

$I_C = \beta I_B$




$\beta >> 1 \rightarrow (\beta+1) I_B \approx \beta I_B$

$V_{BE}$

(1) $\rightarrow$ $10 I_B + 0.7 + 2 I_E = 5 - 0$

$\Rightarrow 10 I_B + 2 \times (100 I_B) = 5 - 0.7$

$\Rightarrow I_B (210) = 4.3$

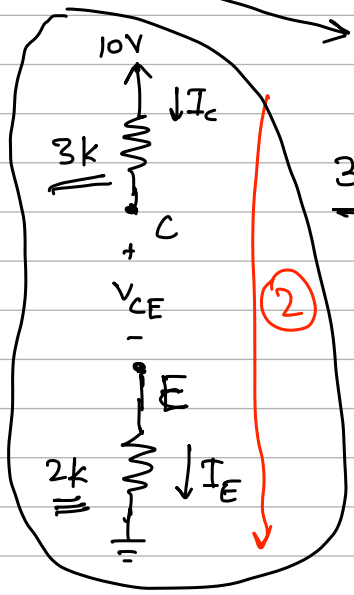
$I_B = 0.0205$ mA ✓

✓ $I_C = \beta I_B = 2.05$ mA

✓ $I_E = I_B + I_C = 2.07$ mA

$V_C = 10 - 3 I_C$

$V_E = 2 I_E$



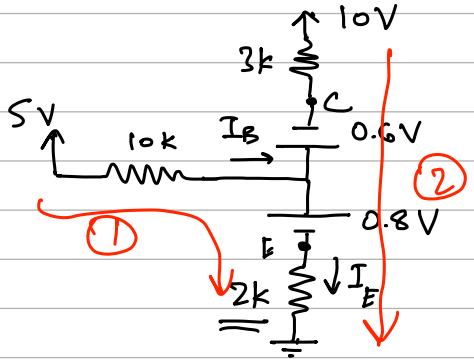

$3 I_C + V_{CE} + 2 I_E = 10 - 0$

$V_{CE} = 10 - 6.15 - 4.141$

$= 10 - 10.3$

$= -0.3 < 0.2$ V

Assumption wrong ! Assume saturation




(1) $\rightarrow$ $10 I_B + 0.8 + 2 I_E = 5 - 0$

but, $I_C \neq \beta I_B$, $I_E \neq (\beta+1) I_B$

[not in active]

$V_{CE} = 0.2$

in saturation, $I_C \approx I_E$ : (2) $\rightarrow$ $3 I_C - 0.6 + 0.8 + 2 I_E = 10 - 0$ ($\approx I_C$)

✗ wrong (approx.)

$\Rightarrow 5 I_C = 10 - 0.2$

$I_C = 1.96$ mA

(1) $\rightarrow$ $10 I_B = 5 - 0.8 - 2 \times 1.96 = 0.28 \Rightarrow I_B = 0.028$ mA

Without approx.:

$0I_C$ $\quad (5 - 0.8)$

① → $10 I_B + 0 I_C + 2 I_E = 4.2$

② → $0 I_B + 3 I_C + 2 I_E = 9.8$

$-I_B + I_C + I_E = 0$

③ $I_E = I_B + I_C$

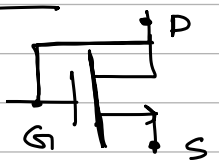
⇒ $-I_B - I_C + I_E = 0$

3 eq^n's, 3 unknowns → solve

$I_B = 0.03$ mA , $I_C = 1.95$ mA , $I_E = 1.98$ mA

Mosfet:

(Neaman) - n-MOSFET ch3 → circuit solve using MAS




↳ we use $k = k_n' \frac{W}{L}$ ,

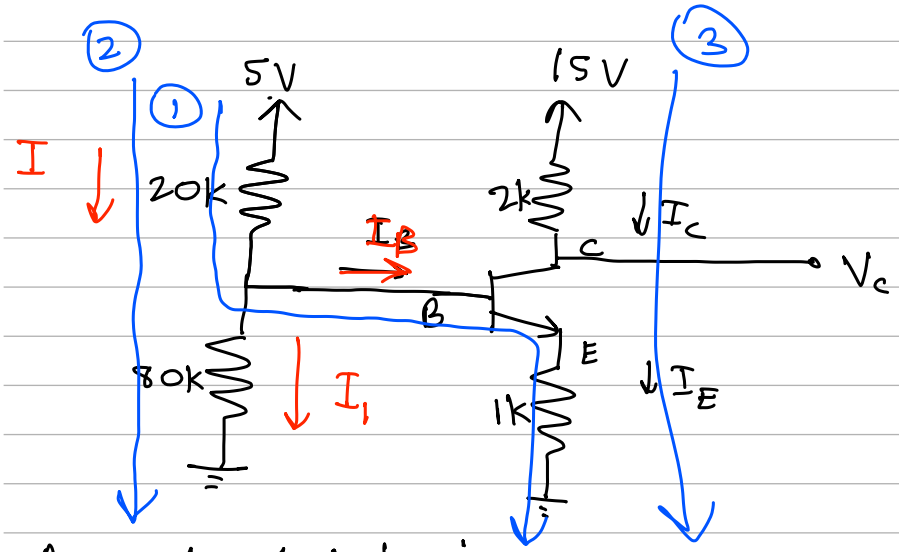
neaman uses $k = \frac{k_n'}{2} \frac{W}{L}$

Sedra Smith — n MOSFET ch5

BJT:

(Neaman) - BJT ch5 → circuit solve using MAS
(npn)

Sedra — BJT ch6




$\beta = 100$

Assumption : Active : $V_{BE} = 0.7V$, $I_C = \beta I_B$, $I_E = (\beta+1) I_B$

$= 101 I_B$

$I = I_1 + I_B$

① → $20I + \overbrace{0.7}^{V_{BE}} + 1I_E = 5 - 0$

② → $20I + 80I_1 = 5 - 0$

→ $20I_1 + 20I_B + 101I_B = 4.3$

$\Rightarrow 20I_1 + 121 I_B = 4.3$ — ①

$20I_1 + 20I_B + 80I_1 = 5$

$\Rightarrow 100 I_1 + 20 I_B = 5$ — ②

Solving ①, ② → $I_1 = 0.044$ mA, $I_B = 0.028$ mA

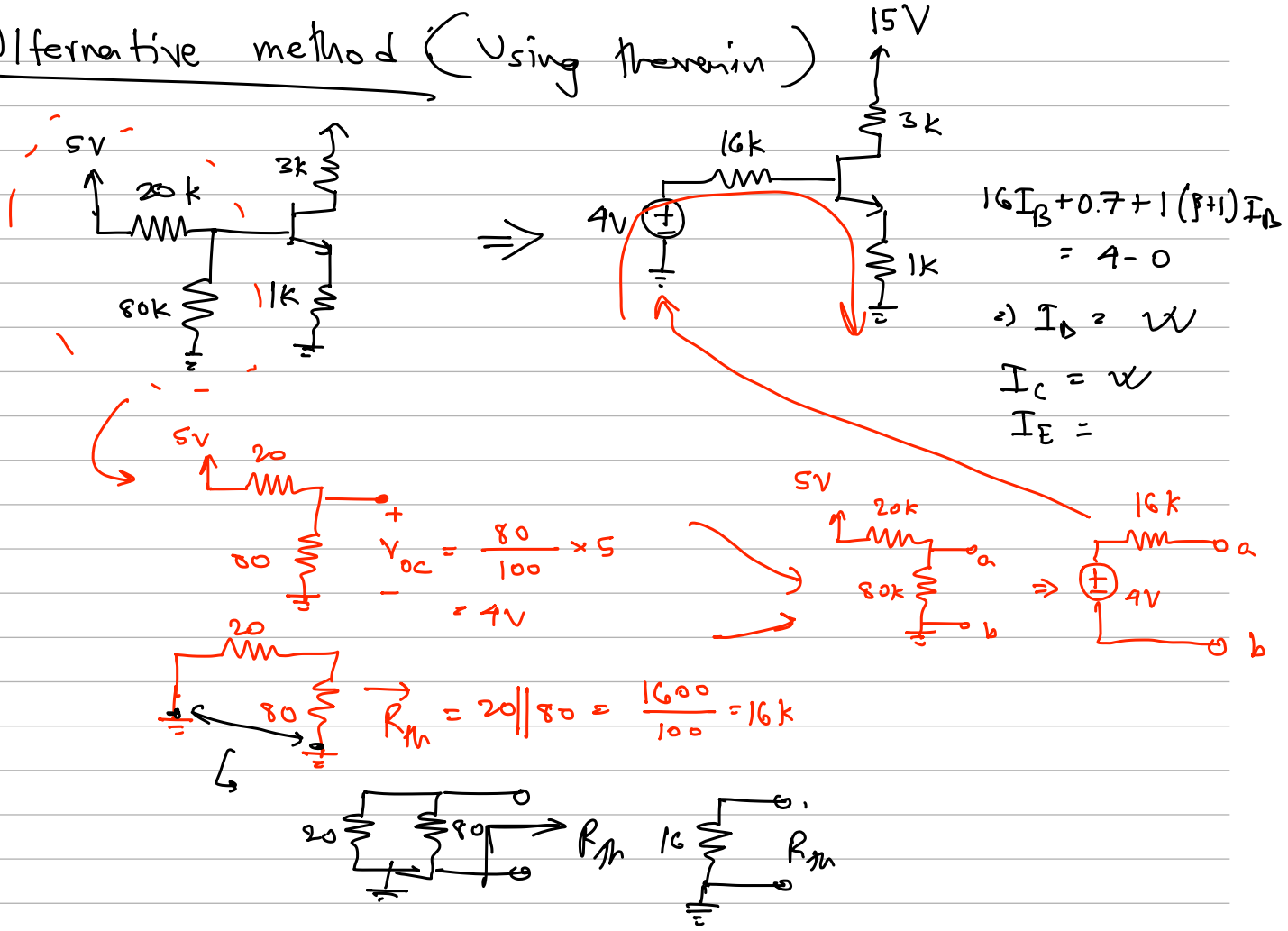
$I_C = 100 I_B = 2.8$ mA, $I_E = 101 I_B = 2.8028$ mA

③ → $2I_C + V_{CE} + 1I_E = 15 - 0$

$\Rightarrow V_{CE} = 15 - (2 \times 2.8) - (1 \times 2.8028)$

$= 6.59 V > 0.2V$

# Alternative method (Using thevenin)




Final syllabus:

1. Zener
2. MOSFET
3. BJT